

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT : JAVA PROGRAMMING

Practical No : 11

Aim 11A : Write a Servlet that accepts a User Name from a HTML form and stores it as a cookie. Write another Servlet that returns the value of this cookie and displays it.

CODE :

pract11A_index.html :

```
<!DOCTYPE html>
<html>
<head>
    <title>Cookie Example</title>
</head>
<body>

    <h2>Enter User Name</h2>

    <form action="storeCookie" method="post">
        User Name:
        <input type="text" name="username" required>
        <br><br>

        <input type="submit" value="Submit">
    </form>

</body>
</html>
```

pract11A_readCookie.java :

```
import java.io.IOException;
import java.io.PrintWriter;
import java.net.URLDecoder;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.Cookie;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/readCookie")
public class pract11A_readCookie extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
```

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```
Cookie cookies[] = request.getCookies();
String username = "Cookie Not Found";

if (cookies != null) {
    for (Cookie cookie : cookies) {

        if (cookie.getName().equals("username")) {
            username = URLDecoder.decode(cookie.getValue(), "UTF-8");
            break;
        }
    }
}

response.setContentType("text/html");

PrintWriter out = response.getWriter();

out.println("<h2>Stored User Name</h2>");
out.println("<h3>User Name: " + username + "</h3>");
}
}
```

pract11A_storeCookie.java :

```
import java.io.IOException;
import java.io.PrintWriter;
import java.net.URLEncoder;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.Cookie;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/storeCookie")
public class pract11A_storeCookie extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String username = request.getParameter("username");

        String encodedUsername = URLEncoder.encode(username, "UTF-8");
```

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```
Cookie cookie = new Cookie("username", encodedUsername);
cookie.setMaxAge(60 * 60 * 24);

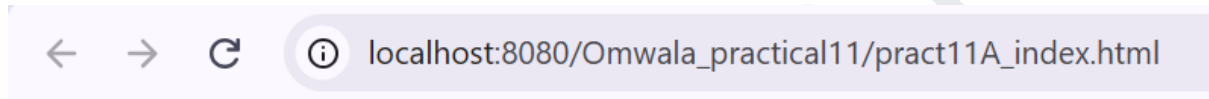
response.addCookie(cookie);

response.setContentType("text/html");

PrintWriter out = response.getWriter();

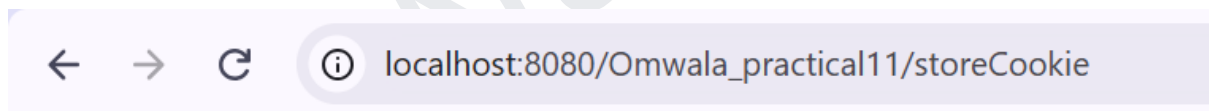
out.println("<h2>User Name stored successfully!</h2>");
out.println("<a href='readCookie'>Read Cookie</a>");
}
```

OUTPUT :



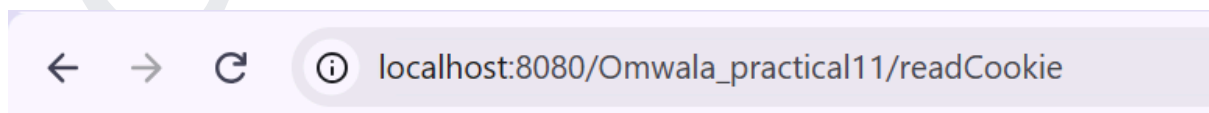
Enter User Name

User Name:



User Name stored successfully!

[Read Cookie](#)



Stored User Name

User Name: Om Wala

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Aim 11B : Write a Servlet that displays the names and values of the cookie stored on the client.

CODE :

pract11B_displayCookies.java :

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.Cookie;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/displayCookies")
public class pract11B_displayCookie extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        out.println("<html>");
        out.println("<head>");
        out.println("<title>Display Cookies</title>");
        out.println("</head>");
        out.println("<body>");

        out.println("<h2>Cookies Stored on Client</h2>");

        Cookie cookies[] = request.getCookies();

        if (cookies != null) {

            out.println("<table border='1' cellpadding='10'>");
            out.println("<tr>");
            out.println("<th>Cookie Name</th>");
            out.println("<th>Cookie Value</th>");
            out.println("</tr>");
```

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```
for (Cookie cookie : cookies) {

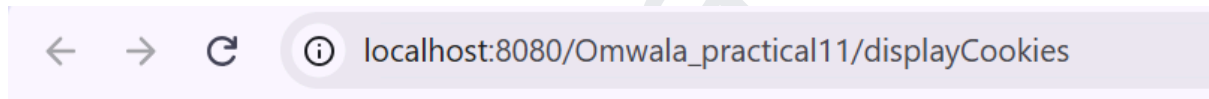
    out.println("<tr>");
    out.println("<td>" + cookie.getName() + "</td>");
    out.println("<td>" + cookie.getValue() + "</td>");
    out.println("</tr>");
}

out.println("</table>");

} else {
    out.println("<h3>No cookies found.</h3>");
}

out.println("</body>");
out.println("</html>");
}
}
```

OUTPUT :



Cookies Stored on Client

Cookie Name	Cookie Value
username	Om+Wala

Aim 11C : Write a Servlet that accepts a User Name from a HTML form and stores it as a session variable. Write another Servlet that returns the value of this session variable and displays it

CODE :

pract11C_index.html :

```
<!DOCTYPE html>
<html>
<head>
    <title>Session Example</title>
</head>
<body>
```

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<h2>Enter User Name</h2>

<form action="storeSession" method="post">

User Name:

<input type="text" name="username" required>

<input type="submit" value="Submit">

</form>

</body>

</html>

pract11C_readSession.java :

import java.io.IOException;

import java.io.PrintWriter;

import javax.servlet.ServletException;

import javax.servlet.annotation.WebServlet;

import javax.servlet.http.HttpServlet;

import javax.servlet.http.HttpServletRequest;

import javax.servlet.http.HttpServletResponse;

import javax.servlet.http.HttpSession;

@WebServlet("/readSession")

public class pract11C_readSession extends HttpServlet {

protected void doGet(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {

HttpSession session = request.getSession();

String username = (String) session.getAttribute("username");

response.setContentType("text/html");

PrintWriter out = response.getWriter();

out.println("<h2>Stored User Name</h2>");

out.println("<h3>User Name: " + username + "</h3>");

}

}

pract11C_storeSession.java :

import java.io.IOException;

import java.io.PrintWriter;

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```
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

@WebServlet("/storeSession")
public class pract11C_storeSession extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String username = request.getParameter("username");

        HttpSession session = request.getSession();

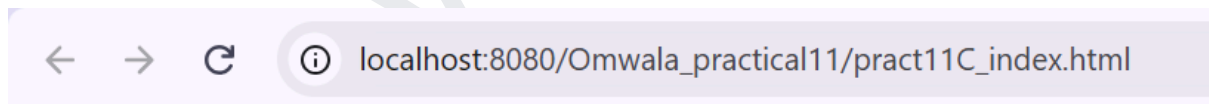
        session.setAttribute("username", username);

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        out.println("<h2>User Name stored in session successfully!</h2>");
        out.println("<a href='readSession'>Read Session</a>");
    }
}
```

OUTPUT :



Enter User Name

User Name:

← → ↻ ⓘ localhost:8080/Omwala_practical11/storeSession

User Name stored in session successfully!

[Read Session](#)

← → ↻ ⓘ localhost:8080/Omwala_practical11/readSession

Stored User Name

User Name: Om Wala